

Aircraft and Rocket Propulsion : Mission Analysis

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1 Mission Specifications

The total mission range is 6,900 km. The distance is distributed across the following flight phases:

Phase	Distance (km)
Taxi	5
Takeoff	20
Climb	150
Cruise	6,500
Descent	200
Landing & Taxi	25
Total	6,900

2 Assumptions and Constants

- **Aircraft:** Gulfstream G650 (MTOW = 45,178 kg)
- **Climb Gradient:** Single-engine gradient is 2.4%, resulting in $\sin \theta = 0.024$.
- **TSFC:** An average value of 0.35 kg/(N · hr) is assumed.
- **TSFC (SI Units):** $TSFC = \frac{0.35}{3600} = 9.72 \times 10^{-5} \text{ kg}/(\text{N} \cdot \text{s})$.

3 Atmospheric Density Calculations

Density is calculated using the Ideal Gas Law: $\rho = \frac{P}{RT}$.

1. **Sea Level (38°C = 311 K):**

$$\rho = \frac{101,325}{287 \times 311} = 1.135 \text{ kg/m}^3$$

2. **Climb/Descent (Average Altitude 7.75 km):**

$$T = 288.15 - (6.5 \times 7.75) = 237.8 \text{ K}$$

$$P = 101,325 \times \left(\frac{237.8}{288.15} \right)^{5.256} = 38,742 \text{ Pa}$$

$$\rho = \frac{38,742}{287 \times 237.8} = 0.568 \text{ kg/m}^3$$

3. Cruise Altitude (15.5 km):

$$T = 216.65 \text{ K}, P = 11,597 \text{ Pa}$$

$$\rho = \frac{11,597}{287 \times 216.65} = 0.1865 \text{ kg/m}^3$$

4 Phase-by-Phase Calculations

4.1 Taxi (Out)

- **Time:** $t = \frac{5000 \text{ m}}{10 \text{ m/s}} = 500 \text{ s}$
- **Fuel Burned:** $F_{fuel_{taxi}} = (9.72 \times 10^{-5}) \times 7500 \text{ N} \times 500 \text{ s} = 365 \text{ kg}$
- **Weight at End:** $45,178 - 365 = 44,813 \text{ kg}$

4.2 Takeoff

- **Stall Speed:** $V_{stall} = \sqrt{\frac{2 \times 443,197}{1.135 \times 119.2 \times 1.6}} = 63.96 \text{ m/s}$
- **Takeoff Speed:** $V_{takeoff} = 1.2 \times 63.96 \approx 77 \text{ m/s}$
- **Time:** $t = \frac{20000}{77} \approx 260 \text{ s}$
- **Lift Coefficient (C_L):** $C_L = \frac{2 \times 439,622}{1.135 \times 77^2 \times 119.2} = 1.097$
- **Induced Drag (C_{di}):** $C_{di} = \frac{1.097^2}{\pi \times 7.75 \times 0.85} = 0.0581$
- **Thrust ($T/W = 0.3$):** $T = 0.3 \times 439,622 = 131,887 \text{ N}$
- **Fuel Burned:** $(9.72 \times 10^{-5}) \times 131,887 \times 260 \approx 3,332 \text{ kg}$

4.3 Climb

- **Time:** $t = \frac{150,000}{150} = 1,000 \text{ s}$
- **Lift Coefficient (C_L):** $C_L = \frac{2 \times 41,483 \times 9.81}{0.568 \times 150^2 \times 119.2} = 0.534$
- **Climb Thrust:** $T = D + W \sin \theta = 25,750 + (407,149 \times 0.024) = 35,522 \text{ N}$
- **Fuel Burned:** $(9.72 \times 10^{-5}) \times 35,522 \times 1,000 \approx 3,450 \text{ kg}$

4.4 Cruise

- **Velocity:** $V_{cruise} = 0.85 \times 295 \approx 250 \text{ m/s}$
- **Time:** $t = \frac{6,500,000}{250} = 26,000 \text{ s}$
- **Lift Coefficient (C_L):** $C_L = \frac{2 \times 38,033 \times 9.81}{0.1865 \times 250^2 \times 119.2} = 0.537$
- **Thrust ($T=D$):** $T = \frac{1}{2} \times 0.1865 \times 250^2 \times 119.2 \times (0.02 + 0.01393) = 23,563 \text{ N}$
- **Fuel Burned:** $(9.72 \times 10^{-5}) \times 23,563 \times 26,000 = 5,960 \text{ kg}$

4.5 Descent and Landing

- **Descent:** $t = \frac{200,000}{180} = 1,111 \text{ s}$, $Fuel = (9.72 \times 10^{-5}) \times 5000 \times 1111 = 540 \text{ kg}$
- **Landing:** $t = \frac{25,000}{40} = 625 \text{ s}$, $Fuel = (9.72 \times 10^{-5}) \times 7500 \times 625 = 456 \text{ kg}$

5 Summary Mission Profile

Phase	W_{start} (kg)	Dist (km)	Thrust (kN)	Time (s)	Fuel (kg)	W_{end} (kg)
Taxi (Out)	45,178	5	7.5	500	365	44,813
Takeoff	44,813	20	132	260	3,330	41,483
Climb	41,483	150	35.5	1,000	3,450	38,033
Cruise	38,033	6500	23.6	26,000	5,960	32,073
Descent	32,073	200	5	1,111	540	31,533
Land/Taxi	31,533	25	7.5	625	456	31,077
Total		6,900			14,101	

6 Conclusion

The iterative reduction of TSFC to $0.35 \text{ kg}/(\text{N} \cdot \text{hr})$ resulted in a total fuel burn of approximately 14,101 kg. This is consistent with the Gulfstream G650's maximum fuel capacity of 20,049 kg, leaving approximately a 30% fuel reserve.